

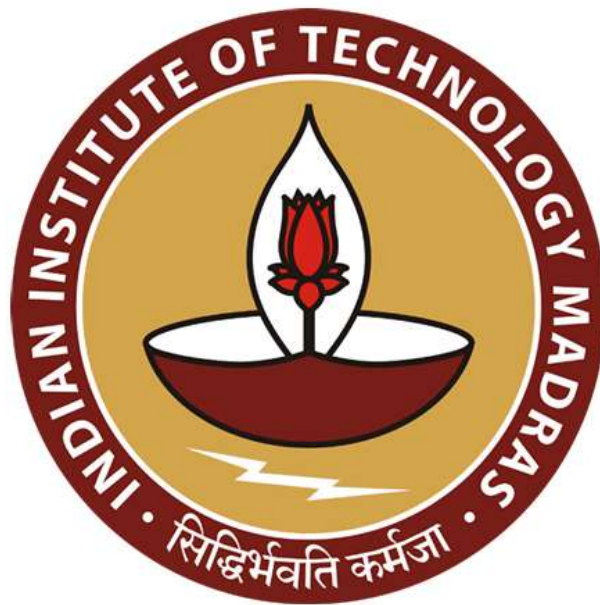
Enhancing Profitability and Cash Flow at a Grocery Store: A Data-Driven Approach to Credit, Financial, and Margin Management

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project “**Enhancing Profitability and Cash Flow at a Grocery Store: A Data-Driven Approach to Credit, Financial, and Margin Management**”. I extend my appreciation to **Faiz General Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

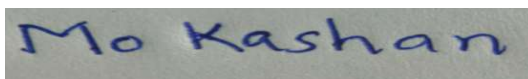
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Mohammad Kashan

Date: 03 September 2025

1 Executive Summary

Faiz General Store is a small B2C neighborhood grocery shop in Prayagraj, catering to local customers with essential daily products like rice, flour, oil, pulses, and packaged goods.

The key business challenges identified are: (i) managing customer credit and ensuring timely repayment, (ii) analyzing revenue contribution and profit margin variation across categories, and (iii) evaluating overall financial health through day-wise revenue, profit, and cash vs. credit sales trends. These challenges directly impact cash flow, profitability, and customer trust.

To address these issues, a data-driven approach will be adopted. Customer transaction history will be analyzed to design a simple credit rating system, supported by rule-based methods and basic classification techniques. Category-wise revenue and margin analysis will be carried out to identify high-performing and low-performing categories, providing actionable insights for pricing and stocking decisions. A financial health assessment framework will be developed using day-wise summary data, incorporating metrics such as revenue growth trends, profit stability, and liquidity indicators.

The expected outcome is improved credit risk management, deeper revenue and margin insights, and enhanced financial health monitoring, thereby strengthening the long-term financial sustainability of the grocery store.

2 Organization Background

Faiz General Store, established in 1995, is a small-scale, family-owned neighborhood grocery shop based in Prayagraj that caters to the daily needs of local households. Operating as a Business-to-Consumer (B2C) setup, it remains open every day, including Sundays, from 7:00 AM to 10:00 PM. The store offers a wide range of products, including rice, flour, oil, pulses, snacks, beverages, and household essentials.

The shop serves a diverse customer base comprising regular, occasional, and walk-in customers. Regular customers often purchase on credit, which creates both opportunities and risks for the business. The store's primary objective is to balance customer satisfaction with sustainable profitability.

3 Problem Statement

3.1 Customer Credit Rating: To identify and classify customers based on repayment behavior into low, medium, and high credit risk categories.

3.2 Category-wise Revenue & Profit Margin Analysis: To analyze revenue contribution and profit margin variation within categories and compare profitability across categories, highlighting high and low performing segments.

3.3 Profit & Financial Health Analysis: To assess overall financial health by examining revenue, profit stability, and cash versus credit sales trends using day-wise summary data.

4 Background of the Problem

Small retail businesses like Faiz General Store typically operate without structured systems for financial or operational management, relying instead on manual methods and intuition. This approach leads to three recurring and interconnected problems that impact cash flow, profitability, and long-term stability:

Unmanaged Credit Sales: While essential for maintaining customer loyalty, credit sales create significant liquidity challenges due to delayed repayments and the risk of default. The absence of a system to track customer creditworthiness or repayment history leaves the business vulnerable to payment defaults and cash shortages.

Limited Visibility into Revenue & Margins: The store lacks a structured analysis of revenue contribution and profit margin variation across categories. As a result, decision-making is often driven by sales volume rather than profitability, leading to missed opportunities to focus on high-margin categories or control losses from low-margin ones.

Weak Financial Health Monitoring: With no systematic evaluation of daily revenue, profit, or the share of cash vs. credit sales, it becomes difficult to measure financial stability. This lack of visibility hinders the ability to assess liquidity, track performance trends, or prepare for risks such as prolonged credit delays or seasonal demand fluctuations.

5 Problem Solving Approach

The project addresses three key challenges of Faiz General Store: customer credit risk, category-wise revenue & profit margin, and overall financial health analysis.

5a. Methods

Credit Rating System:

- Features such as delay days, payment status, credit frequency, and outstanding dues will be extracted.
- A rule-based scoring system will provide initial classification.
- Logistic Regression will be applied (if needed) as a baseline machine learning model to classify customers into Good, Average, and Risky categories.
- **Justification:** These methods are simple, interpretable, and effective for limited datasets, making them practical for small business decision-making.

Category-wise Revenue & Profit Margin Analysis

- $\text{Revenue} = \text{Quantity Sold} \times \text{Sale Price}$
- $\text{Profit} = \text{Revenue} - (\text{Purchase Price} \times \text{Quantity Sold})$
- $\text{Profit Margin} = \text{Profit} / \text{Revenue}$
- Use descriptive statistics and visualizations (bar/line charts) to compare category performance.
- **Justification:** Provides clear insights into high vs. low performing categories.

Profit & Financial Health Analysis

- Day-wise and month-wise revenue and profit trends will be analyzed to assess business stability.
- Key ratios, such as cash-to-credit sales ratio, gross profit margin, and profit volatility, will be calculated to evaluate liquidity, profitability, and risk exposure.
- Visualizations will highlight profit patterns and anomalies across time.
- **Justification:** This approach provides deeper, actionable insights into overall store performance, enabling data-driven decisions for cash flow management, credit control and strategic planning.

5b. Data Collection

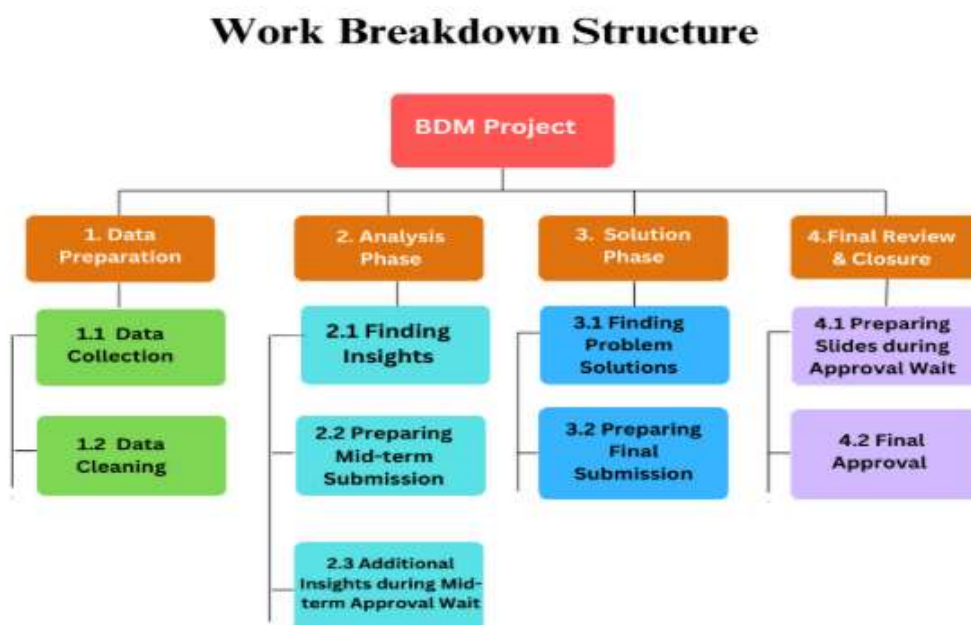
- Data will be collected from Sep 2024 – Aug 2025(Day-wise summary) and from March 2025 – Aug 2025(Detailed transaction level) for analysis.
- **Sources:** Sales invoices, purchase bills, customer credit records, repayment logs, and daily sales summaries.
- **Justification:** 6 months of transaction data is sufficient for micro-level analysis (credit and category margins) since in daily-needs grocery businesses, customer purchase cycles are short and repayment behavior emerges quickly, while the one-year summary ensures robust evaluation of overall financial health.

5c. Analysis Tools

- Google Sheet, Python (Pandas, Matplotlib, Seaborn, Scikit-learn), SQL.
- **Justification:** These tools are cost-effective, widely used in academia and industry. SQL enables efficient data extraction, Python allows scalable preprocessing and modeling, while Google Sheets ensures accessibility and clarity for the shop owner.

6 Expected Timeline

6.1 Work Breakdown Structure



. Figure 1 Work Breakdown Structure

6.2 Gantt Chart

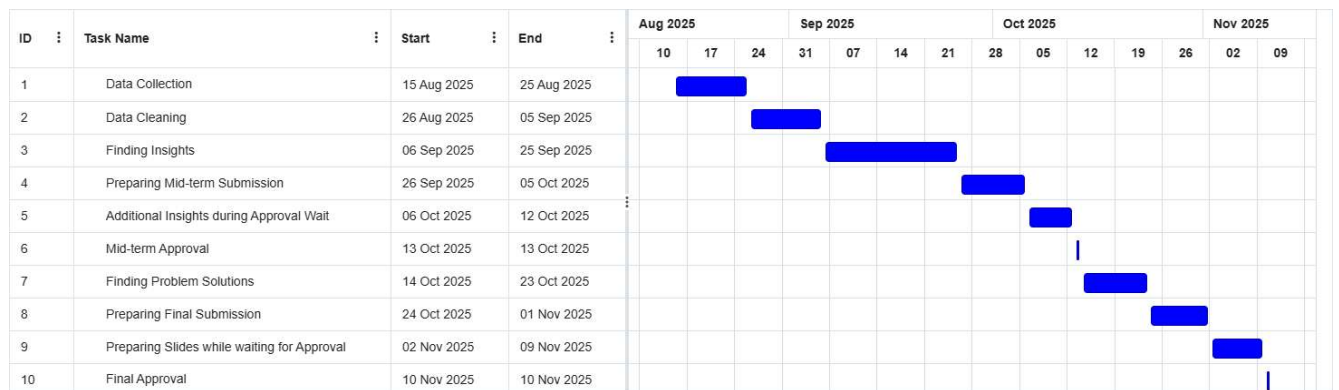


Figure 2 Expected timeline for completion of project

7 Expected Outcomes

The project will deliver:

- A data-driven credit rating scorecard (classifying customers as 'Good', 'Average', 'Risky') to reduce payment defaults and inform credit decisions.
- Detailed insights into category-wise revenue contribution and profit margin variation, identifying high and low performing categories to guide pricing, promotions, and stocking strategies.
- A comprehensive financial health analysis framework using day-wise summary data, highlighting revenue trends, profit stability, and cash versus credit sales patterns to improve liquidity and long-term stability.
- Customer segmentation combines credit behavior and purchase frequency, enabling the store to identify loyal buyers, manage risky customers, and design targeted promotions.

These deliverables will empower Faiz General Store with data-driven decision-making, improving financial stability, operational efficiency, long-term profitability, and customer trust.